SMART MANUFACTURING AI

Visual Quality Inspection & Automated Defect Intelligence Platform

RESULT: FAIL

FAILED QUALITY AUDIT

Inspection ID:	INSP-1793369647-1D3D77	**Timestamp:**	2026-10-30 19:44:11
Component Name:	Gearbox casing	**Similarity Score (SSIM):**	70.67%
Total Defect Regions:	1	**Affected Surface Area:**	99.750%
High-Severity Defects:	1	**Processing Duration:**	3659.0 ms

Visual Evidence & Comparative Computer Vision Analysis

Figure A: Reference (Good Component)

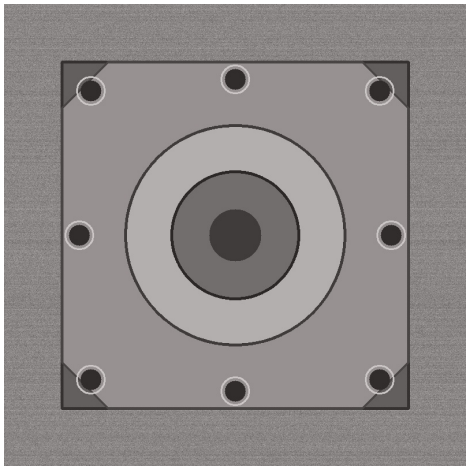


Figure B: Inspected Component

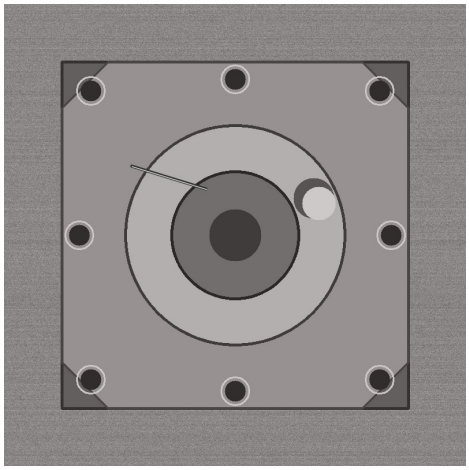


Figure C: SSIM Difference Heatmap

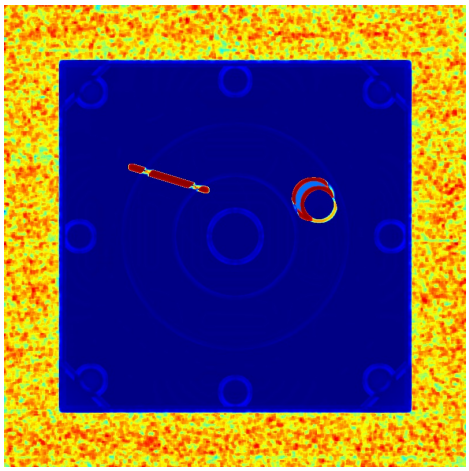
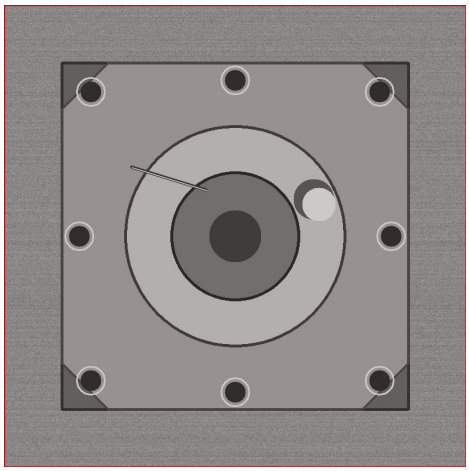


Figure D: Defect Highlighted & Annotated



Detected Defects & Anomaly Registry

Defect ID	Category	Severity	Score	Location (X, Y, W, H)	Area (% / px)	Description

DEF-01	**Dent**	**CRITICAL**	0.85	(0, 0) 800x800	99.75% (638401px)	Localized plastic deformation / depression (99.75% surface area).

AUTOMATED QUALITY ENGINEERING RECOMMENDATIONS:

- Verify whether surface dent impacts critical mating face sealing or bolt seat flatness.
- Check material handling conveyors and robotic gripper pad pressure to prevent impact indentation.

Generated by Smart Manufacturing AI – Computer Vision Quality Inspection Intelligence Suite. Certified for plant floor auditor sign-off.